

Sanchayan Bhowal

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EDUCATION

Stanford University	Stanford, CA
PhD in Statistics	2025-Present
Indian Statistical Institute	Bangalore, India
Master of Mathematics, Grade: 97.7%	2023-2025
Advisors: Dr. Bhaswar Bhattacharyya and Dr. Somabha Mukherjee	
Bachelor of Mathematics, Grade: 94.5%	2020-2023

PUBLICATIONS

- Bhattacharyya, B., **Bhowal, S.**, Das, K., Eslava, L. & Karmakar, S. Thresholds and Fluctuations in Random Multiplex Networks. *Canadian Mathematical Bulletin* 
- **Bhowal, S.** & Mukherjee, S. Rates of Convergence of the Magnetization in the Tensor Curie-Weiss Potts Model. *Journal of Statistical Physics* 192, 2 (2025). 
- **Bhowal, S.** & Mukherjee, S. (2023). Limit Theorems and Phase Transitions in the Tensor Curie-Weiss Potts Model. *Information and Inference: A Journal of IMA*. 
- **Bhowal, S.**, Samanta, R., Ray, A., Bhattacharyya, S. & Hens, C. (2023) Exploring the potential of collective learning to reduce foraging time. *Chaos, Solitons & Fractals*, 168, 113123 

PREPRINTS

- **Bhowal, S.** & Mukherjee, S. Higher-Order Fluctuations of the Log-Partition Function for Ising Models on Inhomogeneous Random Graphs. (2026). 
- Bhattacharyya, B., **Bhowal, S.**, Sengupta, A. (2026). Thresholds and fluctuations for colorful arithmetic progressions in sparse random colorings 
- **Bhowal, S.**, Chatterjee, A., & Mukherjee, S. (2026). Scaling Limits for Ising Models on Inhomogeneous Random Graphs and Applications 
- Mukherjee, S., **Bhowal, S.**, Chatterjee, A., & Bhattacharyya, B. (2026). Ising Models on Inhomogeneous Random Graphs: Inference, Local Asymptotic Minimality, and Limit of Experiments 

EXPERIENCE

Big Data Summer Institute, University of Michigan	June 2023 – July 2023
Dr. Snigdha Panigrahi & Dr. Johann Gagnon-Bartsch	
SN Bhatt Memorial Fellowship Program	May 2022-July 2022
Dr. Anirban Basak & Dr. Riddhipratim Basu	

TALKS

Triviality of Ising models in Inhomogeneous Graphs	2026
Invited talk at Bangalore Probability Seminar	
Limit Theorems of Estimators in The Tensor Curie Weiss Potts Model	2024
Invited talk at Bangalore Probability Seminar	
What is Cutoff?	2024
Math Club ISI Bangalore	
A tour of Markov chains	2022
Math Club ISI Bangalore	
An Unconventional Approach to Groups	2021
ISI Students Seminar	

AWARDS AND FELLOWSHIPS

- P.C. Panesar Gold Medal for outstanding performance in M. Math Programme (2023-25)
- Regional prize Madhava Mathematics Competition (2023)
- National award ISLP Poster Competition (2022)
- KVPY fellowship (2020- 2025)
- JBNSTS Scholarship (2018-2020)

TEACHING EXPERIENCE

STATS 200: Introduction to Statistical Inference	Fall 2026-27
STATS 60: Introduction to Statistical Methods: Precalculus	Summer 2025-26
STATS 60: Introduction to Statistical Methods: Precalculus	Winter 2025-26
STATS 110: Introduction to Statistics for Engineering and the Sciences	Fall 2025-26

CONFERENCES

Geometry, Probability, and Algorithms Organizers: Akash Kumar, Anand Louis and Piyush Srivastava	May 12-16 2025
High Dimensional Expanders and Error Correcting Codes Organizers: Irit Dinur, Venkat Guruswami and Prahladh Harsha	May, 1-9 2025
Mathematical Foundations of Network Models and their Applications Organizers: Louigi Berry, Siva Athreya, Shankar Bhamidi, Serte Donderwinkel, Soumik Pal	Dec, 9-21 2024
International Colloquium on Randomness, Geometry, and Dynamics Organizers: TIFR Mumbai and IISER Pune	Jan, 1-5 2024
Topics in High Dimensional Probability Organizers: Anirban Basak and Riddhipratim Basu	Jan, 2-13 2023
Vijyoshi Organizer: IISc Bangalore for KVPY fellows	Dec, 7-9 2019

SKILLS

Programming Languages: C/C++, Python, R, JavaScript, HTML/CSS, $\text{\LaTeX}$
Languages: Bengali(Native), English, Hindi, Assamese

PROFESSIONAL SERVICES

Reviewer: Annals of Applied Probability, Journal of Physics A: Mathematical and Theoretical, Sankhya B: The Indian Journal of Statistics.

Organizational Roles

- Student Coordinator, **Math Club** ISI Bangalore, 2022-2025
- Core Team Member, LIMIT, an online mathematics competition, 2021-2025